Anas Mhana

Senior Software Engineer

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Professional Summary

Senior Software Engineer with 11 years of experience at the intersection of high-performance C++ development and Computer Vision. Proven track record in designing and optimizing real-time embedded software for ADAS and communication systems at industry leaders like Bosch, DENSO, and Rohde & Schwarz. Eager to apply expert skills in modern C++ and machine learning to solve complex challenges in autonomous systems.

Technical Skills

Languages	C++17, Python, Matlab
Libraries	OpenCV, Pytorch, STL
Tools	Git, CMake, Docker, Bitbucket, MISRA, Scrum, Visual Studio, Linux, Pycharm, VCode

Professional Experience

2023–2025 **Software Development Engineer (via ALTEN GmbH)**, Rohde & Schwarz GmbH, München, Germany.

- Maintained and enhanced high-performance C++ applications for mission-critical radio communication testers.
- Implemented robust software design patterns to improve code maintainability and long-term scalability.
- Improved software reliability by identifying and resolving complex bugs in the existing codebase.
- Optimized core algorithms, significantly reducing processing time and enhancing system performance.

2022–2023 **Real-Time Embedded Software Engineer (via ALTEN GmbH)**, Bosch GmbH, Lindau, Germany.

- Engineered and developed C++17 real-time embedded software components for multi-container, multi-layer appliance systems.
- Led the redesign of the appliance power management logic and the pairing mechanism with cloud services.
- Ensured system robustness through rigorous bug-fixing, endurance testing, and comprehensive documentation.

2019–2021 **Computer Vision Development Engineer**, DENSO ADAS Engineering Services GmbH, Lindau, Germany.

- Developed and trained deep learning models for depth completion and 3D reconstruction tasks.
- Researched and applied novel computer vision technologies to estimate vehicle depth perception.
- Developed real-time visual odometry algorithms and corresponding unit tests in C++.
- Validated algorithm performance by defining evaluation metrics and experimenting with real-world data.

2016–2019 **Computer Vision Development Engineer**, ADASENS AUTOMOTIVE GMBH, Lindau, Germany.

- Developed robust C++ software features and unit tests for an offline mono-camera calibration system.
- Designed and built a validation tool to ensure calibration accuracy and reliability.
- Created an optical self-diagnosis system by modeling and training traditional machine learning models.

2013–2014 **Software Engineer**, MMIS FOR INFORMATION TECHNOLOGY, Lebanon.

2011–2013 **Algorithm and Data Structure Lab Instructor Assistant**, DAMASCUS UNIVERSITY, Syria.

2010–2011 **Programmer**, EBLA GMBH, Syria.

Training & Certifications

2022 **Artificial Intelligence Engineering**, *DIGETHIC*, München, Germany.

Education

2014–2016 **Master of Science in Vision & Robotics (VIBOT)**, *Université de Bourgogne*, France.

2011–2012 **Master of Robotics Programming Engineering**, *Damascus University*, Syria.

2005–2010 **Computers and Automation Engineering**, *Damascus University*, Syria.

Academic Projects & Research

Master's Thesis *Visual based localization in large-scale environments.* Developed a visual localization system using HOG features and linear SVM classifiers.

2015–2016 **Autonomous Robotics with TurtleBot2**, UNIVERSITÉ DE BOURGOGNE, France.

- Developed ROS scripts for autonomous navigation (SLAM) and implemented a state machine for object-recognition-based task execution.

Computer Vision Algorithms Package, UNIVERSITÉ DE BOURGOGNE, France.

- Implemented a suite of classic computer vision algorithms (e.g., feature extraction, filtering) with a Qt GUI.

Languages

Arabic (Mother Tongue), English (Fluent), German (Advanced)